LAB2REPORT

UNIX / C Programming Exercises

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Module: Programming Methodology

Exercise 1: Hello World

Source file: `first.c`

This program prints a simple Hello World message to the terminal.

Source Code

```c
#include <stdio.h>

	int main(void)
	{
	printf("Hello World \n"); return 0;
}
```

Exercise 2: Pattern Output

Source file: `exercise1.c`

This program prints a welcome message followed by an X-shaped pattern.

Source Code

```c
#include <stdio.h>

        int main(void)
        {
        printf("Welcome to C Programming!\n");
        printf("X X\n");
        printf(" X X\n"); printf(" X\n");
          printf(" X X\n");
          printf("X           X\n");
          return 0;
          }
```

Exercise 3: Missing Heights

Source file: missing_heights.c

This program reads three known heights and an average, then calculates the two equal missing heights. It uses the average equation:

$$\frac{sumoffiveheights}{_____} = average\ 5$$

```c
#include <stdio.h>

int main(void)
{
        float perimeter, length, width;

        printf("Enter the perimeter: ");
        scanf("%f", &perimeter);
```

```
9
10          length = perimeter / 3.5;
```

Source Code

```c
#include <stdio.h>

int main(void)
{
        float h1, h2, h3, average;
        float missing_height;

        printf("Enter three known heights: ");
        scanf("%f %f %f", &h1, &h2, &h3);

        printf("Enter the average height: ");
        scanf("%f", &average);

        missing_height = (5 * average - (h1 + h2 + h3)) / 2;

        printf("First missing height: %.2f\n", missing_height);
        printf("Second missing height: %.2f\n", missing_height);
        return 0;
}
```

Exercise 4: Fence Dimensions

Source file: fence_demo.c

This program reads a perimeter and calculates the length and width using the given relationship:

$$length = \frac{perimeter}{3.5}, \qquad width = \frac{3}{4} \times length$$

Source Code

```
11          width = (3.0 / 4.0) * length;

12
13          printf("Length: %.2f\n", length);
14          printf("Width: %.2f\n", width);

15          return 0;

16  }
```

```
agais@ASHAN-PC:/mnt/c/Users/agais/OneDrive/Desktop/IT23632578/Lab_2$ ./first
.o
hello world
agais@ASHAN-PC:/mnt/c/Users/agais/OneDrive/Desktop/IT23632578/Lab_2$ vim exe
rcise1.c
agais@ASHAN-PC:/mnt/c/Users/agais/OneDrive/Desktop/IT23632578/Lab_2$ gcc exe
rcise1.c -o exercise1.o
agais@ASHAN-PC:/mnt/c/Users/agais/OneDrive/Desktop/IT23632578/Lab_2$ ./exerc
ise1.c
./exercise1.c: line 3: syntax error near unexpected token `('
./exercise1.c: line 3: `int main(void)'
agais@ASHAN-PC:/mnt/c/Users/agais/OneDrive/Desktop/IT23632578/Lab_2$ ./exerc
ise1.o
Welcome to C Programming!
   X   X
    X X
     X
    X X
   X   X
```

```
agais@ASHAN-PC:/mnt/c/Users/agais/OneDrive/Desktop/IT23632578/Lab_2$ gcc mis
sing_heights.c -o missing_heights.o
agais@ASHAN-PC:/mnt/c/Users/agais/OneDrive/Desktop/IT23632578/Lab_2$ ./missi
ng_heights.o
Enter three known heights: 122
32
12
Enter the average height: 23
First missing height: -25.50
Second missing height: -25.50
agais@ASHAN-PC:/mnt/c/Users/agais/OneDrive/Desktop/IT23632578/Lab_2$ ./missi
ng_heights.o
Enter three known heights: 150
160
170
Enter the average height: 160
First missing height: 160.00
Second missing height: 160.00
agais@ASHAN-PC:/mnt/c/Users/agais/OneDrive/Desktop/IT23632578/Lab_2$ vim fen
ce_demo.c
agais@ASHAN-PC:/mnt/c/Users/agais/OneDrive/Desktop/IT23632578/Lab_2$ gcc fen
ce_demo.c -o fence_demo.o
agais@ASHAN-PC:/mnt/c/Users/agais/OneDrive/Desktop/IT23632578/Lab_2$ ./fence
_demo.c
./fence_demo.c: line 3: syntax error near unexpected token `('
./fence_demo.c: line 3: `int main(void)'
agais@ASHAN-PC:/mnt/c/Users/agais/OneDrive/Desktop/IT23632578/Lab_2$ ./fence
_demo.o
Enter the perimeter: 120
Length: 34.29
Width: 25.71
agais@ASHAN-PC:/mnt/c/Users/agais/OneDrive/Desktop/IT23632578/Lab_2$
```